Enumeration of Polygon-Free Mosaics

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Abstract

Hong and Oh introduced a model for multiple ring polymers in physics in which an $m \times n$ retangular lattice is constructed from a selection of 7 distinct tiles. These lattices are called *mosaics*. The authors provide bounds on a subset of these mosaics that both contain polygons and have all other tiles that are not part of a polygon set to the blank tile. We introduce and enumerate mosaics with the related property of not containing a polygon, which we call polygon-free mosaics.

1 Mosaics

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, blindfolded. At your disposal is an unending supply of 7 distinct types of tiles. These tiles, diagrammed in Figure 1, are composed of unit squares with dotted lines connecting 2 sides at their midpoints, as well as the "blank" tile T_0.

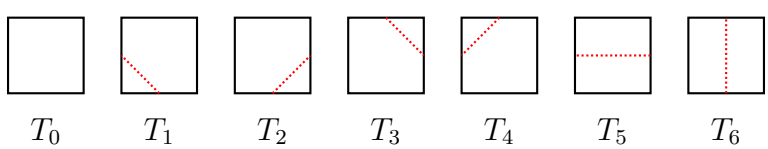
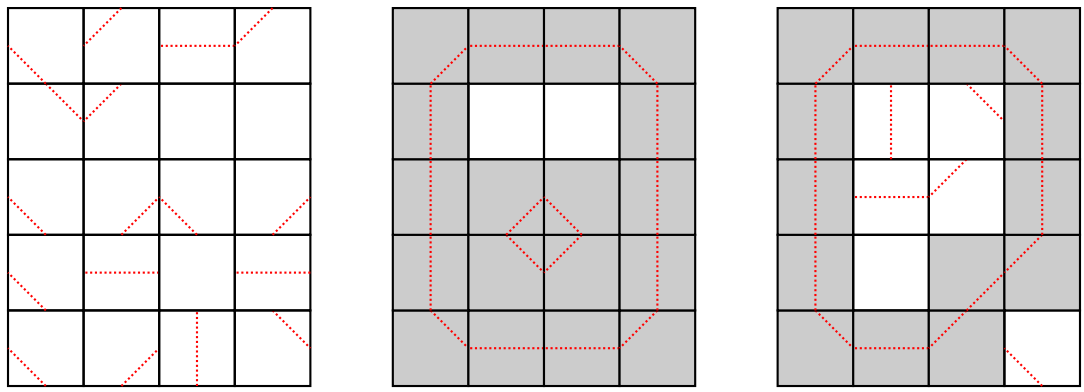


Figure 1: The tile set $\mathbb{T}$

We denote the set of tiles $\mathbb{T} = \{T_0, \ldots, T_6\}$. The task is complete once you place mn randomly selected tiles from $\mathbb{T}$ to cover the floor, after which you remove your blindfold. We are interested in the number of floors that do not contain polygons[1].

We call a fully tiled $m \times n$ floor an (m, n) *mosaic*, where m, n are positive integers. If $m = 5$ and $n = 4$, you may have constructed any of the three mosaics in Figure 2. We highlight the tiles that are part of a polygon in gray.

[1]Polygons are more commonly referred to as "self-avoiding polygons" in the literature to highlight their connection with self-avoiding walks.



(a) A polygon-free mosaic (b) A polygon mosaic (c) A messy polygon mosaic

Figure 2: Examples of mosaics of size $(5, 4)$ made of tiles in $\mathbb{T}$

Definition 1.1. An (m, n) *polygon-free mosaic* is an (m, n) mosaic that does not contain a polygon.

Definition 1.2. An (m, n) *polygon mosaic* is an (m, n) mosaic such that any tile that is not part of a polygon is T_0. Notice this includes the mosaic of all T_0 tiles.

Definition 1.3. An (m, n) *messy polygon mosaic* is an (m, n) mosaic that contains at least one polygon.

Example 1.1. Figure 2a depicts a polygon-free mosaic, Figure 2b depicts a polygon mosaic, and Figure 2c depicts a messy polygon mosaic.

Therefore, let $\mathbb{P}^{(m,n)}$ be the number of (m, n) polygon-free mosaics. Clearly the number of (m, n) messy polygon mosaics is then $7^{mn} - \left|\mathbb{P}^{(m,n)}\right|$. To determine the number of polygon-free mosaics, we begin by defining a related object to mosaics, that being binary lattices.

2 Binary Lattices

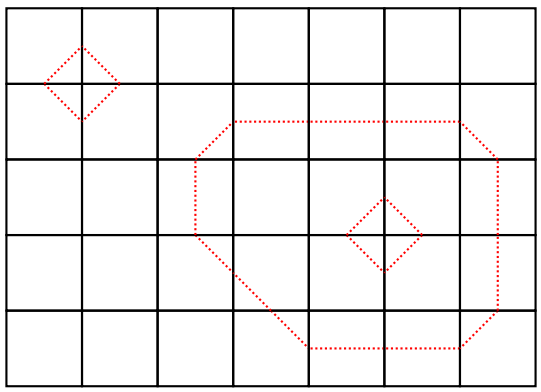
For positive integers m, n, an (m, n) *binary lattice* is a rectangular lattice of $m + 1$ by $n + 1$ vertices, with each vertex labeled 0 or 1. We also define a *framed lattice* to be a binary lattice in which the boundary vertices are labeled 0. An example of a $(5, 7)$ framed lattice is shown on the right of Figure 3.

Definition 2.1. Let a *cell* be a $(1, 1)$ binary lattice.

For convenience, we denote a cell by the 2×2 matrix of its vertex labels. There are sixteen cells:

$$\begin{smallmatrix}0&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\1&0\end{smallmatrix},\ \text{and}\ \begin{smallmatrix}1&1\\1&1\end{smallmatrix}.$$

Definition 2.2. Let $\mathcal{L}$ map (m, n) polygon mosaics to a (m, n) binary lattice with the following rule. If the vertex is surrounded by the red dotted lines of an even number of polygons (including 0 polygons), label it 0. If the vertex is surrounded by the red dotted lines of an odd number of polygons, label it 1. Removing the red dotted lines from the tiles gives the framed binary lattice.



$$\overset{\leftrightarrow}{\mathcal{L}} \quad \begin{array}{ccccccccccccccc}
0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 \\
| & & | & & | & & | & & | & & | & & | & & | \\
0 & — & 1 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 \\
| & & | & & | & & | & & | & & | & & | & & | \\
0 & — & 0 & — & 0 & — & 1 & — & 1 & — & 1 & — & 1 & — & 0 \\
| & & | & & | & & | & & | & & | & & | & & | \\
0 & — & 0 & — & 0 & — & 1 & — & 1 & — & 0 & — & 1 & — & 0 \\
| & & | & & | & & | & & | & & | & & | & & | \\
0 & — & 0 & — & 0 & — & 0 & — & 1 & — & 1 & — & 1 & — & 0 \\
| & & | & & | & & | & & | & & | & & | & & | \\
0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0 & — & 0
\end{array}$$

Figure 3: $\mathcal{L}$ applied to a polygon mosaic

Notice that under $\mathcal{L}$, no tile can map to the cells $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$.

Definition 2.3. Let a *proper lattice* be a binary lattice that does not contain any $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells.

Proposition 2.1. *The map $\mathcal{L}$ is a bijection between (m, n) polygon mosaics and (m, n) proper framed lattices.*

Proof. **TODO** □

3 Counting Polygon Mosaics

We now define a general function v that maps a cell to some integer, then extend v to (m, n) binary lattices by taking the product over v of the mn individual cells in the binary lattice.

For polygon mosaics, we use the following definition for v.

Definition 3.1. Define a map v from cells to integers as follows.

$$\begin{cases} 0 & \text{for cells } \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix} \\ 1 & \text{otherwise} \end{cases}$$

More generally, we define v from binary lattices to integers by taking the product of v applied to each cell in the binary lattice.

Theorem 1. *The number of (m, n) polygon mosaics is*

$$\sum v(\ell),$$

where the sum is over all framed lattices ℓ.

Proof. From Proposition 2.1, there is a bijection between polygon mosaics and framed proper lattices, and so $v(\ell)$ counts one poylgon mosaic for each framed proper lattice ℓ. By Definition 3.1, if ℓ' is a binary lattice that contains either $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, then $v(\ell') = 0$. Therefore, $\sum v(\ell)$ over all framed lattices is equal to the sum over all proper framed lattices, which completes the proof. □

We show that we can more efficiently compute the sum $\sum v(\ell)$ from Theorem 1 using matrices. For matrices, mosaics, and later binary lattices, we number the rows $0, 1, \ldots, m-1$ top to bottom, and columns $0, 1, \ldots, n-1$ left to right.

4 Computing $\sum v(\ell)$ with Matrices

Define $A_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), B_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right), C_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), D_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right)$, and for integers $k \geq 1$,

$$A_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_k \end{pmatrix} \quad B_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_k \end{pmatrix}$$

$$C_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_k \end{pmatrix} \quad D_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_k \end{pmatrix}.$$

We work with general v in this section, and substitute a specific v later.

Definition 4.1. Let the n digit binary representation of the number k be written as $\beta_n(k)$. If n is 0, $\beta_n(k)$ returns the empty string.

We extend our shorthand for cells to $(1, n)$ binary lattice using analagous matrix notation and the definition for $\beta_n(k)$. For example, we write

$$v\begin{pmatrix} 0 & \beta_2(1) & 0 \\ 0 & \beta_2(0) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Proposition 4.1. *For all integers $n \geq 1$, the (i, j)-th entry of*

$$A_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad B_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix},$$
$$C_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad D_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

Proof. We prove by induction. The $n = 1$ case is trivial, as $\beta_0(0)$ is the empty string. We next assume our result for some fixed $n \geq 1$. Since A_n has size $2^{n-1} \times 2^{n-1}$, the $(i, j + 2^{n-1})$ element of A_{n+1} equals $v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ times the (i, j) element of B_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & \beta_{n-1}(i) & 0 \\ 0 & 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_n(i) & 0 \\ 0 & \beta_n(j+2^{n-1}) & 0 \end{pmatrix},$$

as desired. Analagous arguments show that the other blocks in A_{n+1} have the desired values. Similarly, since C_n has size $2^{n-1} \times 2^{n-1}$, the $(i+2^{n-1}, j)$ element of D_{n+1} equals $v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ times the (i,j) element of C_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & 1 & \beta_{n-1}(i) & 0 \\ 1 & 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & \beta_n(i+2^{n-1}) & 0 \\ 1 & \beta_n(j) & 0 \end{pmatrix},$$

again as desired. These arguments must be adapted to prove the desired results for all blocks of B_{n+1}, C_{n+1}, and D_{n+1}, but all the work is similar. □

Theorem 4.2. *For all positive integers m and n, the $(0,0)$ entry of A_n^m equals $\sum v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.*

Proof. Fix n. We first introduce some terminology. For all positive integers m, we say a *height m strip* is an (m,n) binary lattice in which all vertices in the outside columns (i.e., column 0 and column n) are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i < 2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0\ \beta_{n-1}(i)\ 0$. Note that an (m,n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our proposition it suffices to show the following.

Claim. For all positive integers m and nonnegative integers $i, j < 2^{n-1}$, the (i,j) entry of A_n^m equals $\sum v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j.

To prove the claim, first observe that for each i and j, there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j. By Proposition 4.1, v applied to this strip is equal to the (i,j) entry of A_n. This proves our claim when $m = 1$. We continue by induction, so for some $m \geq 2$, assume the claim for $m-1$.

Fix nonnegative integers $i, j < 2^{n-1}$, and consider some nonnegative integer $k < 2^{n-1}$. As in the $m = 1$ case, Proposition 4.1 tells us that the (i,k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k. By our induction hypothesis, the (k,j) entry of A_n^{m-1} equals $\sum v(\ell)$, where the sum is over all height $m-1$ strips ℓ whose first and last rows are determined respectively by k and by j. Consider applying v both to the height 1 strip indicated above and to one of the height $m-1$ strips ℓ in this sum. As in Figure (picture below), by the definition of v, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ.

$$v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}$$

From this we see that the product of the (i,k) entry of A_n and the (k,j) entry of A_n^{m-1} is equal to $\sum v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and last rows are determined respectively by i, by k, and by j. This tells us, in turn, that when we additionally sum over all nonnegative $k < 2^{n-1}$, we can conclude that the (i,j) entry of

$A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j.

Finally, as strips whose first and last rows are determined by $i = j = 0$ are just framed lattices, we get the result. $\square$

Substituting the values for v from Definition 3.1 into the definitions of matrices A_k, B_k, C_k, D_k gives the sum from Theorem 1 by Theorem 4.2. Now that we have enumerated polygon mosaics, we move to the more complicated polygon-free mosaics.

5 Polygon-Free Mosaics

We now redefine v to enumerate polygon-free mosaics.

Definition 5.1. Define a map v from cells to integers as follows.

$$\begin{cases} 7 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix} \\ 0 & \text{for cells } \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix} \\ -1 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix} \\ 1 & \text{otherwise} \end{cases}$$

More generally, we define v from binary lattices to integers by taking the product of v applied to each cell in the binary lattice.

Example 5.1. If we let ℓ be the framed binary lattice on the right of Figure 3, we have $v(\ell) = -7^{11}$, as there are 2 $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cells, 1 $\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}$ cells, 9 $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cells, and 2 $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ cells.

We show our choice of v has useful properties by first considering the magnitude, then the sign. For the magnitude of v, for fixed m, n give an index $1 \leq i \leq N$ to every way one polygon can be formed in an (m, n) mosaic. We remind the reader that a polygon is determined not only by its shape and orientation, but also its location within the mosaic. For example, Figure 4 depicts three distinct polygons in a $(2, 4)$ mosaic, which for reference we call polygons $1, 2$, and 3.

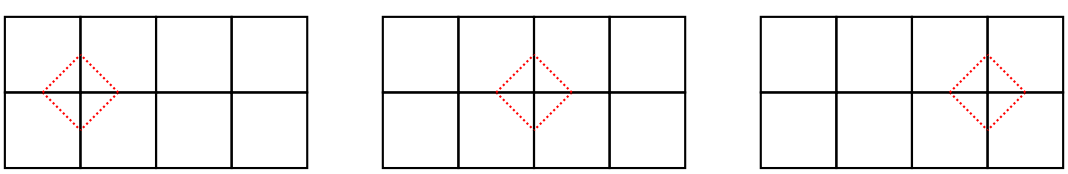


Figure 4: Polygons $1, 2$, and 3

Definition 5.2. Let M_i be the set of (m, n) mosaics that contain polygon i.

We immediately have the number of messy polygon mosaics is

$$7^{mn} - \left|\mathbb{P}^{(m,n)}\right| = \left|\bigcup_{i=1}^{N} M_i\right|. \tag{1}$$

Notice that not every subset of polygons $1 \leq i \leq N$ has a mosaic that contains them all. For example, for the polygons in Figure 4 the second-from-left column contains different tiles in polygon 1 and polygon 2, so no mosaic exists that contain these two polygons. Consequently $M_1 \cap M_2 = \varnothing$.

Proposition 5.1. *Let J be a subset of indices of polygons. If there exists at least one mosaic that contains all polygons in J, we have*

$$\Big|\bigcap_{j\in J} M_j\Big| = \Big|v(\ell)\Big|, \tag{2}$$

where ℓ is the framed binary lattice such that $\ell = f(\mathcal{M})$ where $\mathcal{M}$ is any mosaic that only contains the polygons in J.

Proof. From the definition of M_j, the left hand side of Equation 2 counts the number of mosaics that necessarily contain the polygons in J, and possibly contain other polygons. For the right hand side, if a cell comes from a tile that is part of a polygon, the absolute value of v of the cell is 1. Therefore, $|v(\ell)|$ is just 7 to the number of $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells in ℓ. As all tiles in $\mathbb{T}$ can map to $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$, $|v(\ell)|$ counts all possible tile assignments in all locations that are not part of a polygon in J. Therefore $|v(\ell)|$ counts mosaics that contain the polygons in J. This is the definition of $\bigcap_{i\in S} M_i$, and so the equality holds. □

Definition 5.3. If ℓ is a framed binary lattice that does not contain $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, let $P(\ell)$ to be the number of polygons in the polygon mosaic formed by replacing all $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells with the blank tile T_0, and replacing all other cells with the unique tile that maps to that cell under f.

For the sign of v, we show that though $v(\ell)$ is computed "cell-by-cell", we recover global information about $P(\ell)$.

Proposition 5.2. *If ℓ is a proper framed binary lattice, then*

$$sign(v(\ell)) = (-1)^{P(\ell)}. \tag{3}$$

Proof. For this proof, we use LHS and RHS to abbreviate the left and right hand side of Equation 3. We prove the result by induction. For the base case, construct the $(0, n)$ framed binary lattice ℓ for some $n \geq 0$. As there are no cells in ℓ, from the definition of v the LHS is $\text{sign}(v(\ell)) = 1$. As there no polygons in $g(\ell)$, we have the RHS is 1.

For the induction step, consider any $(m+1, n)$ framed binary strip ℓ' for $m \geq 1$ such that there are no $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells. Similarly, define ℓ to be the framed binary lattice that shares the top m rows of vertices with ℓ', and the bottom most row all being 0. We show that we can construct ℓ' from ℓ with a procedure that preserves Equation 3 with each intermediate step.

Procedure:

Step 1. Add a bottom row to ℓ of $n + 1$ vertices, all labeled 0. This results in a new $(m + 1, n)$ framed binary lattice we denote ℓ_1.

Step 2. Scanning rows $m-1$ and m of ℓ' left to right, if there exists a column of the form $\begin{smallmatrix}1\\1\end{smallmatrix}$, change the associated $\begin{smallmatrix}1\\0\end{smallmatrix}$ column in ℓ_1 to $\begin{smallmatrix}1\\1\end{smallmatrix}$. Completing this scan results in a new framed binary lattice denoted ℓ_2.

Step 3. Again scanning rows $m-1$ and m of ℓ' left to right, if there exists a column of the form $\begin{smallmatrix}0\\1\end{smallmatrix}$, change the associated $\begin{smallmatrix}0\\0\end{smallmatrix}$ column in ℓ_2 to $\begin{smallmatrix}0\\1\end{smallmatrix}$. Completing this scan results in the framed binary lattice ℓ'.

For step 1, only $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cells are added. As $\operatorname{sign}\left(v\left(\begin{smallmatrix}0&0\\0&0\end{smallmatrix}\right)\right)=1$, the LHS is unchanged. For the RHS no new polygons are created in $g(\ell_1)$, so Equation 3 is preserved by step 1.

For steps 2 and 3, we prove the result for each intermediate vertex change as we scan left to right. Furthermore, as each intermediate change only flips a single vertex label from 0 to 1, we only need to consider the 4 cells that share this vertex. We diagram all possible cases for both steps in Figure 5 and Figure 7, where the # symbol indicates the vertex flipping from a 0 to a 1. We know that the bottom most row of vertices must all be 0 by construction.

For step 2, the vertex above the # symbol must be 1 by definition. Also, the vertex right of the # symbol must be 0, as the procedure moves left to right on ℓ_1. Additionally, the $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ cell cannot be created or destroyed, and so Figure 5 depicts all 6 possible cases.

$$\begin{array}{ccccc}0&-&1&-&0\\|&&|&&|\\0&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}\qquad\begin{array}{ccccc}1&-&1&-&0\\|&&|&&|\\0&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}\qquad\begin{array}{ccccc}1&-&1&-&0\\|&&|&&|\\1&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}$$

(a) (b) (c)

$$\begin{array}{ccccc}0&-&1&-&1\\|&&|&&|\\0&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}\qquad\begin{array}{ccccc}1&-&1&-&1\\|&&|&&|\\0&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}\qquad\begin{array}{ccccc}1&-&1&-&1\\|&&|&&|\\1&-&\#&-&0\\|&&|&&|\\0&-&0&-&0\end{array}$$

(d) (e) (f)

Figure 5: Step 2 Cases

As no case in Figure 5 creates or destroys any $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ cells, the LHS sign does not change for any intermediate vertex flip. Similarly, one can check that both before and after the vertex is changed from 0 to 1 that the following is true. Replacing each $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cell with T_0 and replacing all other cells with their unique corresponding tile only forms tiles that must all be part of the same polygon. For example, the corresponding tiles for Case 5e is shown in Figure 6.
Therefore the number of polygons remains the same, and so the RHS is preserved.

For step 3, the vertex above the # symbol must be 0 by definition. As we cannot create or destroy $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells, Figure 7 depicts all 6 possible cases.

We show Equation 3 holds case by case. In Case 7a, the flip add one $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ cell, so the sign

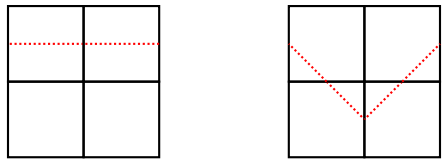


Figure 6: Tile configurations before and after flipping the vertex in Case 5e

$$\begin{array}{ccccc} 0 & — & 0 & — & 0 \\ | & & | & & | \\ 0 & — & \# & — & 0 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array} \qquad \begin{array}{ccccc} 0 & — & 0 & — & 0 \\ | & & | & & | \\ 1 & — & \# & — & 0 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array} \qquad \begin{array}{ccccc} 1 & — & 0 & — & 0 \\ | & & | & & | \\ 1 & — & \# & — & 0 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array}$$

(a) (b) (c)

$$\begin{array}{ccccc} 0 & — & 0 & — & 1 \\ | & & | & & | \\ 0 & — & \# & — & 1 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array} \qquad \begin{array}{ccccc} 0 & — & 0 & — & 1 \\ | & & | & & | \\ 1 & — & \# & — & 1 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array} \qquad \begin{array}{ccccc} 1 & — & 0 & — & 1 \\ | & & | & & | \\ 1 & — & \# & — & 1 \\ | & & | & & | \\ 0 & — & 0 & — & 0 \end{array}$$

(d) (e) (f)

Figure 7: Step 3 Cases

of the LHS changes. As the flip creates a new polygon the sign of the RHS changes. In Case 7b, the flip removes and adds a $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cell, so the sign of the LHS stays the same. As the flip does not create a new polygon, the RHS stays the same. In Case 7c, the flip adds both a $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cell and a $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ cell, so the sign of the LHS stays the same. As the flip does not create a new polygon, the RHS stays the same. In Case 7d, the flip does not add a $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cell or $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ cell, so the sign of the LHS stays the same. As the flip does not create a new polygon, the RHS stays the same. In Case 7e, the flip removes a $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cell, so the sign of the LHS changes. Before the flip, replacing the cells with the appropriate tiles results in a configuration that can either represent portions of one or two polygons, as shown in Figure 8.

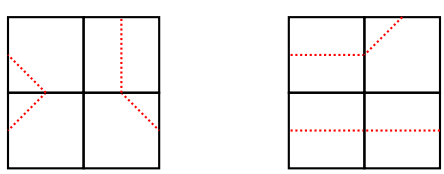


Figure 8: Tile configurations before and after flipping the vertex in Case 7e

If the tiles were part of one polygon before the flip, the tiles after the flip must be part of two distinct polygons. Similarly, if the tiles are were part of two polygons before the flip, the tiles after the flip must be of one polygon. Either way the total number of polygons changes by 1, and so the RHS changes sign. In Case 7f, the flip adds $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$, so the sign of the LHS changes. The same logic from Case 7e applies for the RHS, as the flip either removes

or adds a polygon.

As all cases preserve Equation 3, the $(m+1,n)$ framed binary lattice ℓ' also follows Equation 3. $\square$

Theorem 5.3. *The number of (m,n) mosaics that do not contain a polygon has*

$$\left|\mathbb{P}^{(m,n)}\right| = \sum_{\ell} v(\ell), \tag{4}$$

where the sum is over all (m,n) framed binary lattices.

Proof. We begin by rewriting Equation 1 to

$$\left|\mathbb{P}^{(m,n)}\right| = 7^{mn} - \left|\bigcup_{i=1}^{N} M_i\right|.$$

By the inclusion-exclusion principle, we have

$$-\left|\bigcup_{i=1}^{N} M_i\right| = \sum_{\varnothing \neq J \subseteq \{1,2,\ldots,N\}} (-1)^{|J|} \left|\bigcap_{j\in J} M_j\right|. \tag{5}$$

Consider an individual subset of polygon indices J in the sum in Equation 5 such that $|J| = k \geq 1$. If there exists no mosaic that contains all polygons in J, by the definition of M the term is 0. If at least one mosaic does exist, Propositions 5.1 and 5.2 give the following

$$(-1)^{|J|}\left|\bigcap_{j\in J} M_j\right| = v(\ell),$$

where ℓ is the binary lattice such that $f(\mathcal{M}) = \ell$ for all mosaics $\mathcal{M}$ that contain at least the polygons in J.

Now discard all 0 terms in Equation 5 and group the terms in Equation 5 for all J such that $|J| = k$. From the definition of M, the absolute value of this collection of terms counts the number of mosaics that have at least k polygons, and so

$$\sum_{J\subseteq\{1,2,\ldots,N\},|J|=k} (-1)^{|J|}\left|\bigcap_{j\in J} M_j\right| = \sum_{\{\ell | P(\ell)=k\}} v(\ell).$$

Additionally notice that the binary lattice ℓ^* made up of all $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cells has $v(\ell^*) = 7^{mn}$, and is the only framed binary lattice such that $P(\ell) = 0$. Therefore, we can write

$$\left|\mathbb{P}^{(m,n)}\right| = 7^{mn} - \left|\bigcup_{i=1}^{N} M_i\right| = \sum_{k\geq 0} \sum_{\{\ell | P(\ell)=k\}} v(\ell).$$

Finally, notice that the sum $\sum_\ell v(\ell) = 0$ for the framed binary lattices that contains $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, as $v\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = v\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = 0$. Therefore, summing over all framed binary lattices in Equation 4 gives the desired result. $\square$

As in Section 4, substituting the values of v into matrices A_k, B_k, C_k, D_k gives the sum from Theorem 5.3 by Theorem 4.2. For completeness, we write these substitutions below.

Definition 5.4. For integers $k \geq 1$ let A_k, B_k, C_k, D_k be $2^{k-1} \times 2^{k-1}$ matrices with integer entries, where $A_1 = \begin{pmatrix} 7 \end{pmatrix}, B_1 = \begin{pmatrix} -1 \end{pmatrix}, C_1 = \begin{pmatrix} 1 \end{pmatrix}, D_1 = \begin{pmatrix} 1 \end{pmatrix}$ and

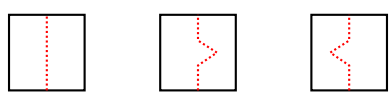
$$A_{k+1} = \begin{pmatrix} 7A_k & B_k \\ C_k & D_k \end{pmatrix} \quad B_{k+1} = \begin{pmatrix} -A_k & B_k \\ 0C_k & D_k \end{pmatrix}$$
$$C_{k+1} = \begin{pmatrix} A_k & 0B_k \\ C_k & D_k \end{pmatrix} \quad D_{k+1} = \begin{pmatrix} A_k & -B_k \\ C_k & 7D_k \end{pmatrix}.$$

Theorem 5.4. *The number of (m, n) polygon-free mosaics $\left|\mathbb{P}^{(m,n)}\right|$ is the $(0, 0)$ entry of A_n^m.*

Finally, we introduce possible extensions to our work, as well as how our work relates to other mosaic enumeration efforts.

6 Extensions and Related Work

Enumerating the number of polygon-free mosaics in certain extensions of the tileset $\mathbb{T}$ is simple to do. For example, if we extend $\mathbb{T}$ to include three types of tiles that connect the top and bottom midpoints like below



one simply needs to adjust the v from Definition 5.1 to have $v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right) = 3$.

In terms of related work, our work can be seen as an extension to Hong and Oh [2] and further generalizing the techniques in Oh et al. [9] to explore a new direction in mosaic enumeration. To begin, Hong and Oh [2] provided bounds on the number of polygon mosaics.

Theorem 6.1 ([2])**.** *The number of (m, n) polygon mosaics for $m, n \geq 3$ is bounded between $2^{m+n-3}\left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3}\left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Theorem 1 nicely compliments this result with a procedure to exactly enumerate polygon mosaics, as well as exactly enumerating the sequence A181245 on the OEIS [4]. It is important to note that contrary to this work, the authors in [2] did not consider the mosaic of all T_0 a polygon mosaic, and so our enumeration differs by 1. We also point out that Theorem 1 is likely known to the authors of [2] and [9], though to our knowledge it has not appeared in print.

In related work, Lomonaco and Kauffman [3] introduced mosaics constructed from a tile set of 11 distinct tiles, of which $\mathbb{T}$ is a subset. The authors were interested in a subset of mosaics which they call *knot mosaics*. Oh et al. [9] enumerated the number of knot mosaics using an analagous method to that of Section 4.

Theorem 6.2 ([9]). *The number of (m, n) knot mosaics for $m, n \geq 2$ is $2\,\|(X_{m-2} + O_{m-2})^{n-2}\|$, where $X_0 = O_0 = \begin{pmatrix} 1 \end{pmatrix}$ and X_{m-2} and O_{m-2} are $2^{m-2} \times 2^{m-2}$ matrices defined as*

$$X_{k+1} = \begin{pmatrix} X_k & O_k \\ O_k & X_k \end{pmatrix} \text{ and } O_{k+1} = \begin{pmatrix} O_k & X_k \\ X_k & 4O_k \end{pmatrix},$$

for $k = 0, 1, \ldots, m-3$. Here $\|N\|$ denotes the sum of elements of matrix N.

Oh and colleagues go beyond enumeration by bounding the growth rate of knot mosaics [6, 8, 1], and Oh further adapts techniques similar to Section 4 method to solve problems in monomer and dimer tilings [5, 7]. Related ideas were independently used to enumerate the number of rectangular partitions of a rectangle in [10] for the sequence A182275 on the OEIS [4].

We point out it is not possible to extend our results naively to the Lomonaco and Kauffman [3] tile set to enumerate the number of *knot-free mosaics*. This is specifically due to the deductions for Case 7e and Case 7f in the proof of Theorem 5.2, as knots allow for additional subcases in which the number of knots does not change.